Sprint II Progress Review: Core Functionality, Integration and Initial Testing

CAMR: A Lightweight External Memory Engine to Bridge the Capability Gap Between On-Device and Cloud Language Models

Student	**Ibrahim Sudheysi Ali**	**Admission no.**	_____
Supervisor	_____	Review date	5 November 2026
Repository	github.com/Satasdin/CAMR (branch camr-build)	Project type	AI/ML (experimental) + application

1. Sprint goal

Move from the prepared environment and data to a working core: a frozen local model that remembers what it is told as its own keys and values in rooms on the SSD, answers from the one room a question needs, and can be compared live against conventional RAG and no memory, with initial tests and evaluation against a baseline.

2. Sprint tasks linked to the objectives

#	Task	Obj.	Status	Evidence	Issue / next step
1	Engine core: remember a moment as the model's notes in a room; recall the closest room already read	ii, iii	**Done**	v2/camr_engine; Fig. 4.4–4.7; unit tests	—
2	Tiers and sleep under RAM and disk budgets; 8-bit rooms	iii	**Done**	T-04, T-05; §5.8 speed, RAM, storage	Room files still large on the SSD
3	Assistant: routing, Wikipedia research, files (PDF, image, video), self-revision, topic shelves	ii	**Done**	T-18 to T-22	Auto shelf sometimes picks the wrong shelf
4	Native Mac app: Chat only, code answered in Chat, live tensor view	iii	**Done**	§5.4 screenshots; /Applications/CAMR.app	Some LaTeX shown untypeset
5	Memory modes CAMR · RAG · Off and Compare, every figure measured	iv	**Done**	T-23; Fig. Compare	Live gap small (50–80 ms) at one room
6	Long Haul harness: CAMR vs model alone vs Claude, graded by rule	iv	**Done**	§5.8; records 2026-11-04	—
7	Component comparisons: desk, reach, notes vs words, reader size, wording	v	**Done**	§5.8	—
8	Multimodal memory and Code (café websites) benchmarks	iv	**Done**	§5.8 media, code	Small samples
9	Held-out CAMR vs RAG head-to-head on the Long Haul	iv, v	**In progress**	Pre-registered (E-61); harness built	Stopped before the held-out run; Sprint III
10	Automated tests in CI	iv	**Done**	GitHub Actions on Apple silicon: 652 pass; sandboxed-code grader tests run on the Mac	803 pass locally
11	Chapter 4 diagrams and Chapter 5 updated to the built system	all	**Done**	CAMR_Thesis.docx, 130 pp.	Student to update Word fields
12	Deployment preparation: website, release download	iii	**In progress**	site/ (local)	Hosting and a release URL in Sprint III

3. Core workflow to demonstrate (input → processing → output)

 Tell: Remember "Iten Traders raised their minimum order to KES 52,000." → judged against earlier notes, filed on a shelf, read once by the model, its notes written to a room on the SSD.

- **Ask:** "What is Iten Traders' minimum order now?" → routed to memory, the closest room laid on the desk already read → answer with its source, note, tier and first-word time.
- **Compare:** "Who is the landlord of the Karen branch?" → CAMR and RAG both answer Patrick Chacha (first word 257 ms against 304 ms; 458 against 749 tokens read as text); Off does not know.
- **Tensors (⌘T):** the room read flows into the model's 36 layers; in RAG mode the chunks enter as text.
- **Note:** press ⌘N for a new chat before switching memory Off, so the conversation does not carry the answer.

4. Initial testing (full record: Chapter 5, Tables of test cases and defects)

Test	Expected	Actual	Verdict
Automated suite (49 files)	All pass	803 passed, 5 skipped	Pass
Memory off (negative)	Model alone does not know a told fact	"I don't know …"	Pass
Second process opens a memory	Refused	ReachBusy	Pass
Foreign web page / no token to the engine	Refused	Security tests pass	Pass
Router on 14 questions	Memory questions to memory	13 of 14; the miss answered honestly	Pass
Notes ≥10× sooner than words (NFR-02)	≥ 10×	4.6×	Fail (kept)

Defects found and retested this sprint include: RAM under-reported (19.2 GB real, fixed to 7.5 GB with identical answers); coding words taken for names; a force-quit app leaving its engine holding the memory's lock (fixed: the engine now exits with the app).

5. Initial evaluation evidence (baseline, metric, held-out data)

Baselines: the same model without memory and Claude alone. Metric: right answers graded by rule, with first word and RAM. Settings were chosen on development seeds; reported runs use the held-out seed 1.

Long Haul (73 questions)	2M tokens	3M	4M	First word	RAM
Qwen3 4B + CAMR	29	29	29	0.2–0.3 s	5.4–7.5 GB
Qwen3 4B alone	11	12	13	0.3–0.4 s	15.2 GB
Claude Sonnet 5.5 alone	32	13	10	13–70 s	cloud

Against RAG (earlier experiments): level on accuracy (164 vs 166 facts), three times the tool errands (122 vs 40), first word 2–4× sooner. Memory closes knowledge gaps, not reasoning gaps (two-hop questions 0 of 10).

6. One AI suggestion checked

Claude Code reported the engine's peak RAM as 5.7 GB from the records. Checking the process itself showed 19.2 GB: MLX was keeping freed graphics buffers. Accepted only after measuring: the buffer cache was capped at 1 GB, the process fell to 7.5 GB and every verdict stayed identical (decision log E-40, E-41). Rejected: reporting the 5.7 GB figure as the process's RAM.

7. Known limitations and Sprint III plan

- **Limits:** one scripted project; 10 questions per family; one Mac; room files large on the SSD; facts in two rooms not joined; no user study yet.
- **Sprint III:** run the pre-registered CAMR vs RAG head-to-head (E-61); host the website and a release download; fix Auto shelf filing and LaTeX display; a short usability check with representative users; final report and handover.